LINE FOLLOWER ROBOT DIY KIT

User Manual

Learn • Build • Program • Explore

Welcome

Thank you for purchasing the Circuitly.in Line Follower Robot DIY Kit.

This kit is designed to introduce students and hobbyists to the exciting world of robotics, electronics, and Arduino programming. By assembling and programming this robot, you will learn how sensors detect a path and how a robot can autonomously navigate by following a line.

Kit Features

✓ Easy Assembly

✓ Arduino Based

✓ IR Sensor Technology

✓ Differential Drive Control

✓ STEM Learning Project

✓ Beginner Friendly

What is a Line Follower Robot?

A Line Follower Robot is an autonomous robot that follows a predefined path marked by a black line on a white surface.

The robot uses two Infrared (IR) sensors to detect the position of the line and continuously adjusts its movement to stay on track.

Applications include:

- Industrial Automation
- Warehouse Robots

- Automated Guided Vehicles (AGVs)
- Educational Robotics
- Research Projects

Kit Contents:

Arduino UNO Compatible Board	**1**
L298N Motor Driver Module	1
IR Sensor Module	2
Wheels	2
Bo Motor	2
Castor wheel	1
18650 cell	2
18650 cell holder	2
Connector wires	Assorted
Robot chasis	1
Mounting hardware	1kit
Charging module c type	1

Understanding the IR Sensor

The IR sensor consists of:

- Infrared LED (Transmitter)
- Photodiode (Receiver)

White Surface

White surfaces reflect infrared light.

Sensor Output:
HIGH

Black Surface

Black surfaces absorb infrared light.

Sensor Output:
LOW

The Arduino reads these outputs and determines the robot's direction.

Working Principle

The robot uses two IR sensors:

- Left Sensor
- Right Sensor

Case 1: Both Sensors on White Surface

Left = HIGH

Right = HIGH

Action:
Move Forward

Case 2: Left Sensor Detects Black Line

Left = LOW

Right = HIGH

Action:
Turn Left

Case 3: Right Sensor Detects Black Line

Left = HIGH

Right = LOW

Action:
Turn Right

Case 4: Both Sensors Detect Black Line

Left = LOW

Right = LOW

Action:
Stop or Junction Detection

Assembly Instructions

Step 1: Assemble Chassis

1. Attach both motors to the chassis.
2. Install the wheels.
3. Mount the castor wheel.

Step 2: Mount Electronics

1. Fix the Arduino board.
2. Install the L298N motor driver.
3. Mount both IR sensors at the front of the robot.
4. Position sensors approximately 5–10 mm above the ground.

Step 3: Connect Motors

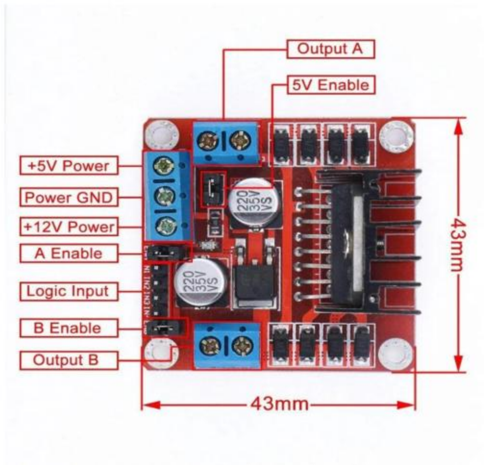
Left Motor → Motor A

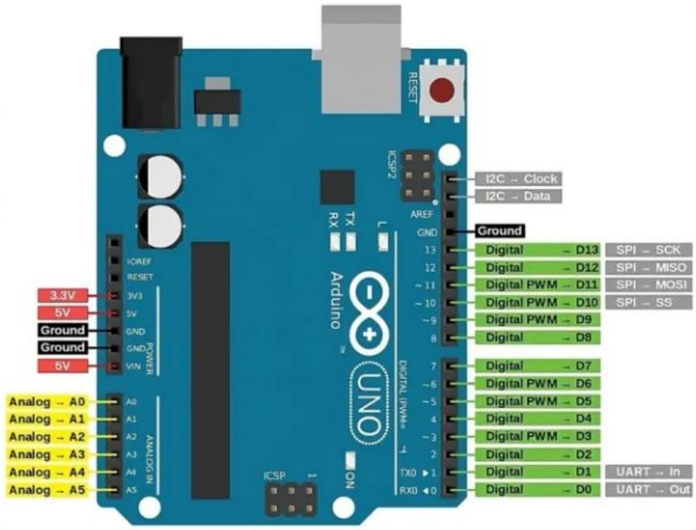
Right Motor → Motor B

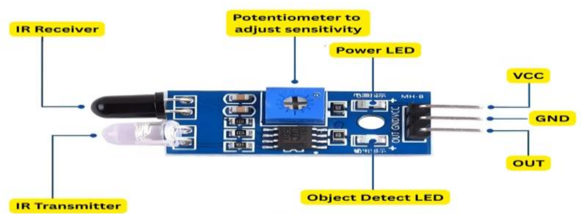
on the L298N module. And remove jumper form ENA and ENB form L298N Module

Pin connection with Arduino

L298N driver Pins	**Arduino Pins**
ENA	5
IN1	9
IN2	10
IN3	7
IN4	8
ENB	6
IR Sensor	**Arduino Pins**
VCC	5V
GND	GND
Do(Right IR)	11
DO(Left IR)	12







Programming the Robot

1. Install Arduino IDE.
2. You can download code from https://github.com/circuitlyin/SmartTrack-kit
3. Watch the step by step guide video https://www.youtube.com/@Circuitlyin
4. Connect Arduino using USB cable.
5. Select:
    - Board: Arduino UNO
    - Correct COM Port
6. Upload the provided Line Follower code.
7. Wait for upload completion.

Sensor Calibration

Each IR module includes a sensitivity adjustment potentiometer.

Calibration Steps

1. Place sensor over white surface.
2. Observe indicator LED.
3. Place sensor over black line.
4. Adjust potentiometer until reliable detection is achieved.

Recommended Line Width:

15 mm – 25 mm

Testing Procedure

1. Create a black line using electrical tape or marker.
2. Place robot on the track.
3. Power ON the robot.
4. Observe movement.
5. Fine-tune sensor sensitivity if necessary.

Troubleshooting

Robot Not Moving

Possible Causes:

- Battery discharged
- Loose wiring
- Program not uploaded

Solution:

- Charge battery
- Verify connections
- Re-upload code

Robot Moves Backward

Possible Cause:

Motor polarity reversed

Solution:

Swap motor wires.

Robot Cannot Follow Line

Possible Causes:

- Sensor height incorrect
- Sensitivity not adjusted
- Poor line contrast

Solution:

- Set sensor height to 5–10 mm
- Calibrate sensors
- Use darker line

Robot Turns Continuously

Possible Causes:

- Sensor misalignment
- Incorrect sensor wiring

Solution:

- Check sensor position
- Verify wiring

Learning Outcomes

After completing this project, students will learn:

- Arduino Programming
- Sensor Interfacing
- Motor Control
- Robotics Fundamentals
- Embedded Systems
- Autonomous Navigation

Support

Circuitly.in

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Thank you for choosing Circuitly.in DIY Learning Kits.

Happy Building! Happy Learning!